RICOCHET

iGaming Skill Game · Original Concept

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1. Executive Summary

Ricochet is an original iGaming skill game in which the player throws a flat stone across a lake. Each bounce off the water surface multiplies the potential payout. The player's challenge is to predict — or influence — how many bounces the stone makes before it sinks.

The game introduces a new visual metaphor for progressive risk: the stone's trajectory and the expanding ripple rings on the water surface create an intuitive, real-time representation of escalating multipliers. Unlike abstract crash or multiplier games, Ricochet grounds the risk mechanic in a universally understood physical phenomenon.

Format: Single-screen HTML5 game · Mobile-first · No registration required for demo

RTP: ~98% across all difficulty modes (house edge 2%)

Session length: 15–90 seconds per round

Volatility: Low (Facile) to Extreme (Expert) depending on mode selected

2. Core Concept & Theme

2.1 Concept

The player places a bet and selects a difficulty mode. A stone is thrown from the bank across the lake. Each time the stone bounces off the surface, a multiplier tier is unlocked. When the stone finally sinks, the round ends and the payout is determined by the number of completed bounces.

The core tension: harder modes have higher per-bounce bust probability, compensated by higher multipliers at each tier. The player chooses difficulty before the throw — a single committed decision, not a continuous cash-out mechanic.

2.2 Unique Selling Points

- Physical intuition: the bounce-and-sink metaphor maps directly to risk/reward without any tutorial
- Visual originality: no existing iGaming title uses stone-skipping as its core visual language
- Clean tension curve: multiplier climbs with each visible ripple ring, creating natural anticipation
- Dual information design: stone trajectory and water ripples simultaneously communicate game state and outcome
- Cross-demographic appeal: the mechanic is universally understood, works for casual and experienced players

2.3 Competitive Positioning

Game	Publisher	Similarity	Ricochet Differentiator
Plinko	Stake Originals	Progressive risk, visual path	Unique physical metaphor — no existing competitor
Crash	Multiple	Multiplier growth, timing	Pre-committed round, no real-time cash-out pressure
Mines	Stake Originals	Progressive unlocking	Continuous animation, more immersive round experience

3. Game Mechanics

3.1 Round Flow

- Player selects difficulty mode (Facile / Moyen / Difficile / Expert)
- Player sets bet amount and confirms
- Single probability roll determines outcome before animation starts (RNG integrity)
- Arc 0 — Throw: stone travels from bank to first water contact. Always present, no sound.
- Arcs 1..n — Bounces: each bounce triggers a sound effect and a ripple ring expanding from the impact point
- Arc n+1 — Sink: stone breaks the surface, water splash animation plays, round ends

3.2 Arc System

The number of bounces (n) is pre-determined at round start via a single probability roll. The animation visualizes the already-determined outcome, maintaining full RNG integrity throughout the round.

Sound design reinforces progression: bounce pitch decreases with each successive impact, creating an audio signal of increasing depth and danger. The final plouf sound is distinct from bounce sounds.

4. Mathematical Model

4.1 Facile — House Edge 2.02%

Bounces	Probability	Multiplier	E[Return]
0 (Loss)	40.0%	×0	0.000
1	32.0%	×1.24	0.397
2	16.0%	×1.65	0.264
3	8.0%	×2.27	0.182
4	3.0%	×3.10	0.093
5	1.0%	×4.44	0.044
TOTAL	100%	Max ×4.44	RTP 97.98%

4.2 Moyen — House Edge 1.60%

Bounces	Probability	Multiplier	E[Return]
0 (Loss)	55.0%	×0	0.000
1	22.0%	×1.35	0.297
2	12.0%	×1.80	0.216
3	6.0%	×2.65	0.159
4	3.0%	×4.20	0.126
5	1.5%	×7.00	0.105
6	0.4%	×14.00	0.056
7	0.1%	×25.00	0.025
TOTAL	100%	Max ×25	RTP 98.40%

4.3 Difficile — House Edge 1.70%

Bounces	Probability	Multiplier	E[Return]
0 (Loss)	70.0%	×0	0.000
1	15.0%	×1.88	0.282

2	8.0%	×2.75	0.220
3	4.0%	×4.15	0.166
4	2.0%	×6.50	0.130
5	0.7%	×10.00	0.070
6	0.2%	×20.00	0.040
7	0.1%	×75.00	0.075
TOTAL	100%	Max ×75	RTP 98.30%

4.4 Expert — House Edge ~2.0%

Bounces	Probability	Multiplier	E[Return]
0 (Loss)	85.0%	×0	0.000
1	8.0%	×3.63	0.290
2	4.0%	×5.18	0.207
3	1.5%	×8.29	0.124
4	0.8%	×12.43	0.099
5	0.4%	×20.72	0.083
6	0.2%	×36.26	0.073
7	0.05%	×62.16	0.031
8	0.03%	×103.59	0.031
9	0.02%	×207.19	0.041
TOTAL	100%	Max ×207.19	RTP 98.04%

4.5 Volatility Summary

Mode	Hit Freq.	Max Win	Volatility	Target Player
Facile	60%	×4.44	Low	Casual / New players
Moyen	45%	×25	Medium	Regular players
Difficile	30%	×75	High	Risk-oriented players
Expert	15%	×207.19	Very High	High-stakes / Bonus hunters

5. Player Psychology

5.1 Anticipation Loop

The stone's visible trajectory during its arc creates a 1–3 second anticipation window per bounce. Unlike instant-result games, Ricochet sustains tension throughout the animation. Each ripple ring expansion is a micro-reward signal: "it survived another bounce."

5.2 Near-Miss Effect

In higher difficulty modes, the stone visibly decelerates and skims lower with each bounce, creating a physical near-miss sensation even in losing rounds. The visual degradation of arc height communicates increasing danger without explicit percentage UI.

5.3 Mode Progression

Four clearly differentiated modes with increasing risk create a natural player progression path. The multiplier gap between modes (×4.44 → ×207) creates strong upward migration incentive as players gain confidence with the mechanic.

6. Audio & Visual Design

6.1 Visual Scene

- Lakeside forest environment with pine and deciduous trees, parallax depth layers
- Dynamic sky with wind-driven clouds (continuous motion in idle)
- Lake with animated water shimmer, lily pads, reed beds
- Fox character on fallen log at lake's edge — watches each throw, idle sway animation
- Stone: angular polygonal silhouette, slate-grey gradient, mineral cracks, hard specular highlight
- Water ripple rings expand from each bounce point and persist until natural dissipation
- Splash: 12 water droplets arc from impact point on sink

6.2 Audio

- Bounce: layered tonal hit, pitch decreases per bounce (danger indicator)
- Sink: physical plouf with reverb tail, distinct from bounce sounds
- Win: ascending arpeggio scaled to win magnitude (bigger win = more notes)

7. Technical Specifications

Specification	Value
Technology	HTML5 · Canvas 2D · Web Audio API
Dependencies	None — fully self-contained single file
File size	~280 KB (includes synthesized audio)
Platform	Desktop · Mobile · Tablet (responsive canvas)
RNG	Math.random() prototype — certifiable backend in production
Frame rate	60 fps, requestAnimationFrame loop
Audio engine	Web Audio API — synthesized, no audio file dependencies

8. Commercial Roadmap

- Backend RNG integration with certified provider (Scientific Games, iTech Labs)
- GLI / BMM certification of math model
- RGS integration (Remote Gaming Server) for multi-operator connectivity
- Ante Bet feature: +25% bet for enhanced bounce probability (Pragmatic Play standard)
- Buy Bonus: direct access to maximum bounce scenario at premium cost
- Localization: EN / FR / DE / ES / PT